

Transfer Your Safety: Learning Transferable Model-Free Safety Filters from a Single Policy to Enhance Safety Across Diverse Tasks

Junjun Xie*, Siru Li*, Shuhao Zhao, Xiaochen Xie, and Liang Hu

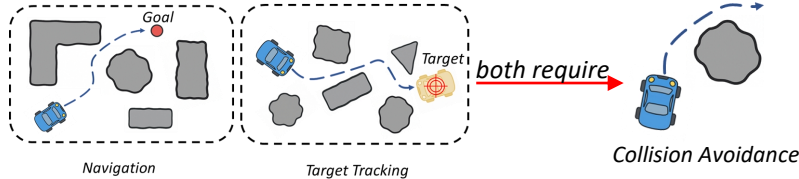
* Equal contribution

Harbin Institute of Technology, Shenzhen



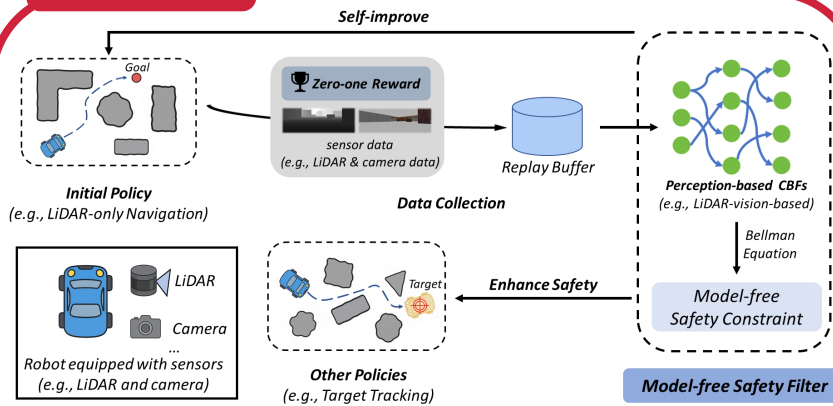
Motivation

- Different tasks share the same safety requirements in similar environments



Given a task-specific policy, can we learn the knowledge of safety and transfer it to other tasks?

Method



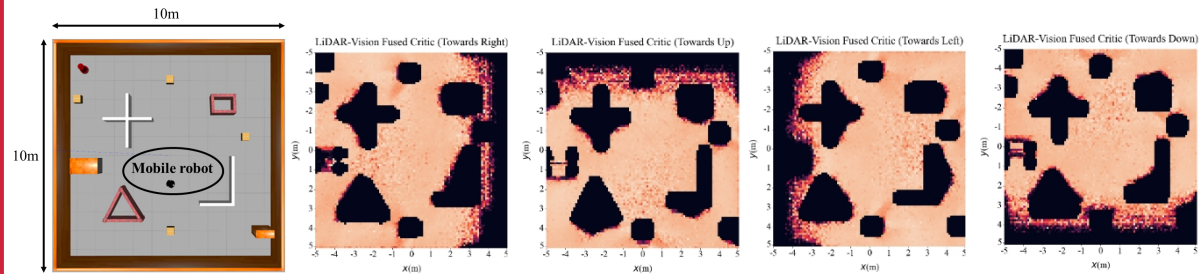
The overall pipeline is as follows:

- Collect data from an initial policy and learn a perception-based Control Barrier Function (CBF) using predefined binary rewards.
- Derive *model-free* action CBF constraints from the Bellman equation without predicting future observations.
- Use the constraints as a safety filter to correct policies for other tasks in similar environments.

Experiments

Perception-based CBF Learning

We pretrained a **LiDAR-based navigation policy** and collected LiDAR data together with camera data unused by the original policy from randomly initialized environments, then trained a CBF based on both **LiDAR and camera observations**.

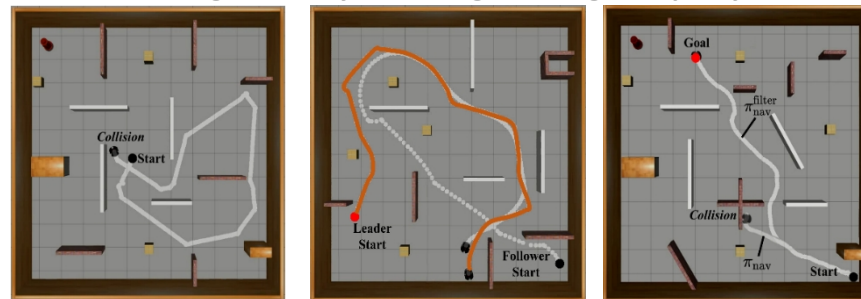


Test Scenario

Visualization of the perception-based CBF
(robot facing right, up, left, and down, respectively).

Cross-task Safety Enhancement & Self-improvement

We derive a model-free safety filter based on the LiDAR-camera-based CBF and apply it to **two obstacle-unaware task policies (cruising and tracking)**, as well as the pretrained navigation policy used for data collection. Results show significant **safety improvements across tasks**, while also enhancing the safety of the original navigation policy.



Cruising

Tracking

Navigation

Cruising

Safety filters	Inputs	Min.	Max.	Average
W/o safety filters	—	35	202	87
Baseline	Single-frame	403	924	530
	Multi-frame	509	1865	987
Ours	Single-frame	462	1007	648
	Multi-frame	524	2089	1116

Tracking

Min. distance	Max. distance	Average	Number of collisions
0.93 m (at step 864)	9.62 m (at step 0)	2.70 m	0

Navigation

	Initial	Filtered
Success rate	13/20(65%)	17/20(85%)